Report: Pre Professional/ Professional Practice

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Vital Jugos SPA
Manufacture, distribution, and commercialization of food and beverage products

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Abstract

This report describes the professional internship experience conducted at Jugos Vital SPA, a strategic subsidiary of the Coca-Cola System in Chile. The internship was carried out within the Maintenance Planning department, under the direct supervision of the area planner. The primary objective was the modernization of asset management through the digitalization of processes and the advanced analysis of forklift fleet data. Key activities included the transition from physical checklists to the Fracttal digital platform, the optimization of shift scheduling and transportation logistics for technical personnel, and the implementation of 5S methodologies. The most significant technical challenge involved the development of a Business Intelligence ecosystem in Power BI. This project required processing and combining multiple data sources to generate critical indicators such as equipment downtime, recurring failure rates, and mean time to resolution. The results allowed for a transition from reactive management to a data-driven strategy, significantly improving operational visibility and decision-making efficiency within the department. This experience provided a comprehensive integration of asset management and logistics skills, which are essential for the professional profile of a civil engineer.

1. Introduction

1.1 Company Description

Jugos Vital SPA is a prominent organization in the Chilean agro-industrial sector, functioning as a vital subsidiary within the Coca-Cola System since its acquisition in 1995. The company specializes in the production, packaging, and distribution of a wide array of food and beverage products, managing high-prestige brands such as Andina del Valle, Kapo, and Powerade. The production facility, which employs between 251 and 500 collaborators, operates under a continuous production regime (24/7) to meet national demand through state-of-the-art aseptic packaging technology and a high-turnover logistics center.

The Maintenance department is a critical pillar of this operation, consisting of a specialized team of 29 workers. This team is supervised by professionals overseeing specific areas such as electrical systems, flexible packaging (Tetra Pak), and utilities. Furthermore, the area relies on a maintenance planner and a technical buyer to ensure the plant's operational continuity and efficiency. In this high-intensity environment, the availability of loading and transport equipment is essential for maintaining the supply chain's integrity, ensuring that all processes adhere to strict international certifications of food safety and excellence.

Figure 1: Organization chart of Vital Jugos Company.

Figure 2: Vital Jugos plant organization chart.

1.2 Department / Area Description

The Maintenance Planning department serves as the strategic core that guarantees the availability and reliability of the plant's industrial assets across all areas, including production, formulation, warehouse, and yard. Led by Matías Valenzuela as the Maintenance Area Planner, the department is responsible for scheduling work orders, managing spare parts, and coordinating exhaustive maintenance shutdowns. The technical team is specialized; while mechanics are assigned to specific production lines (such as bottle or flexible lines), electricians operate transversally across the facility. Furthermore, the area coordinates with external contractors who provide permanent specialized technical support for complex machinery.

The intern's role was centered on supporting the digital transition of the department, acting as a vital link between the workshop and management through the administration of Fracttal software. This involved translating field data from mechanical and electrical technicians into strategic information. The position required constant coordination with the production department, especially during scheduled line stoppages, to ensure that maintenance activities were aligned with operational goals. By managing the logistical needs of both internal staff and contractors, the intern helped minimize production downtime and ensured that the diverse machinery from formulation to bottling operated at peak efficiency.

2. Work Details

2.1 General Activities

The daily responsibilities were centered on optimizing maintenance management and logistics to ensure operational continuity across the plant. This process began with the systematic collection and review of physical operational control checklists from five key areas: production, warehouse, yard, waste management, and formulation. These paper forms, filled out by operators at the start of each shift, served as the primary source of data for identifying critical technical issues such as worn tires, hydraulic leaks, or broken gauges.

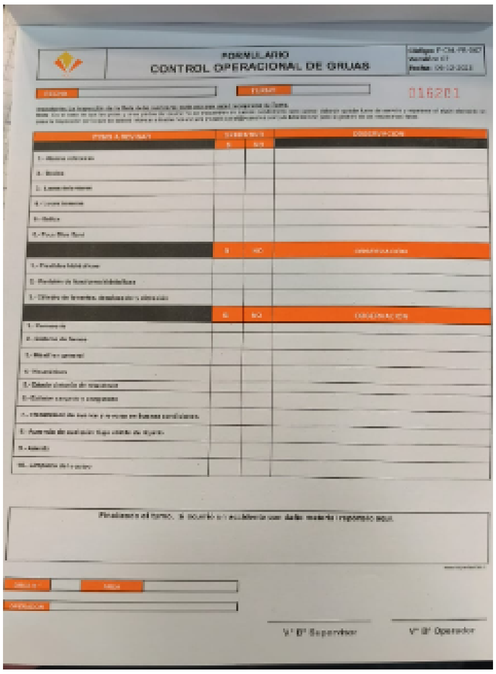
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Image 1: Daily operational control form for forklift fleet.

After reviewing these manual reports, the information was digitized and entered into the Fracttal platform to maintain an orderly record of the fleet's status. I was responsible for assigning preventive work orders at the start of each week or month, following schedules determined by the maintenance planner. This system allowed me to coordinate directly with the mechanical team and external technicians from KREIS to prioritize repairs based on the severity of the failures, ensuring that maintenance resources were allocated efficiently to minimize production downtime.

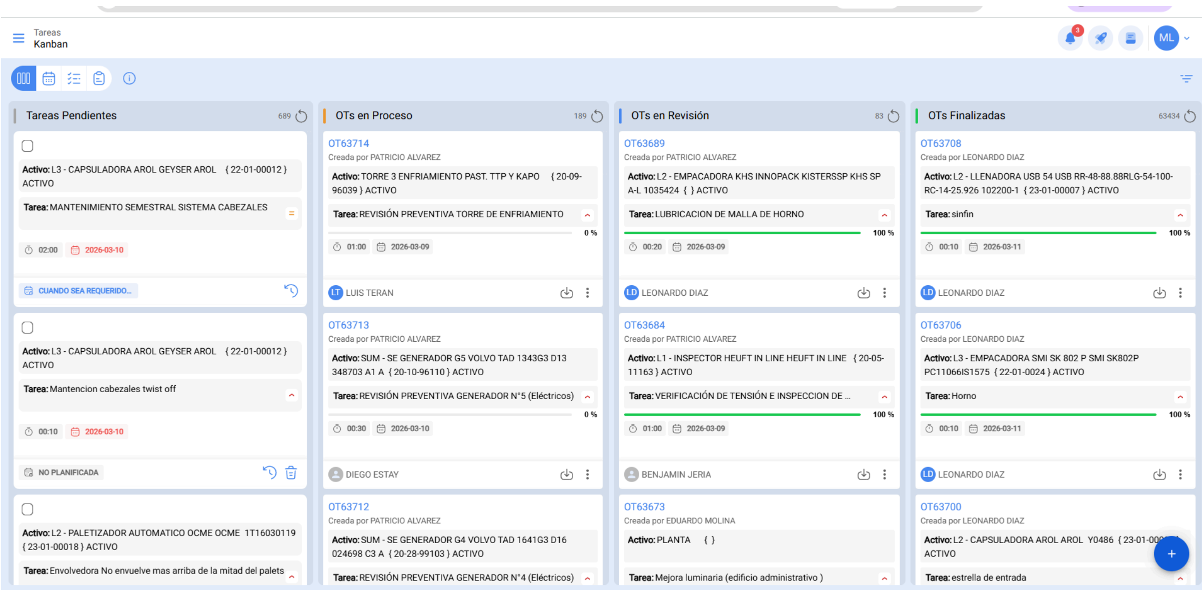

Image 2: Assignment and progress of work orders in FRACTTAL

Furthermore, the assignments were based on failure analysis tools like the bathtub curve and data retrieved from SIMON software. The role also included essential logistical tasks such as managing the weekly schedules for technical personnel across three shifts (A, B, and C). This involved coordinating transportation buses for Shift C to ensure the team arrived and returned safely during night operations. I also collaborated with the Planned Maintenance Pillar by updating the Blackboard Excel file using SIMON data to track production line losses and performance through heat maps. My duties extended to optimizing several Excel templates to automate summaries and labeling, facilitating the search for specific technical data. Other key activities included implementing the 5S methodology, executing a photographic inventory of assets for digital identification, and developing standard operating procedures (SOPs) for equipment cleaning and safety. Finally, I utilized SAP to monitor spare parts inventory and manage procurement orders, verifying technical specifications, prices, and vendors, while physically confirming that incoming items at the warehouse met the required standards.

2.2 Most Challenging Activity

The most demanding technical challenge was the development of an advanced dashboard in Power BI designed to monitor the operational status of the entire forklift fleet. This project represented a significant leap in complexity, as it required the integration of fragmented data sources from five different plant areas: production, warehouse, yard, waste management, and formulation. The primary difficulty lay in the data cleaning and transformation phase; I had to process and merge multiple complex tables to resolve inconsistencies found in manual records. By performing intricate data joins and using DAX functions, I was able to obtain critical metrics that were previously unavailable, such as the total downtime per equipment and the identification of recurring failure patterns within hydraulic and electronic systems.

To ensure the tool provided high strategic value, I implemented a dynamic filtering system that allowed management to analyze the fleet's performance by specific months, plant areas, or the operational status of each unit. A key innovation was the inclusion of a specialized function that tracked and displayed the exact date and time of the last data update, ensuring the reliability and transparency of the information presented. Furthermore, the dashboard enabled the correlation between fault reports and specific work shifts, providing a clear metric for maintenance efficiency and response times. This development was not merely a visualization exercise but a structural improvement that allowed the department to move from reactive management toward a strategy based on statistical evidence. Consequently, this optimized the planning of preventive maintenance and significantly reduced the fleet's overall unavailability, establishing a new standard for data-driven decision-making within the plant.

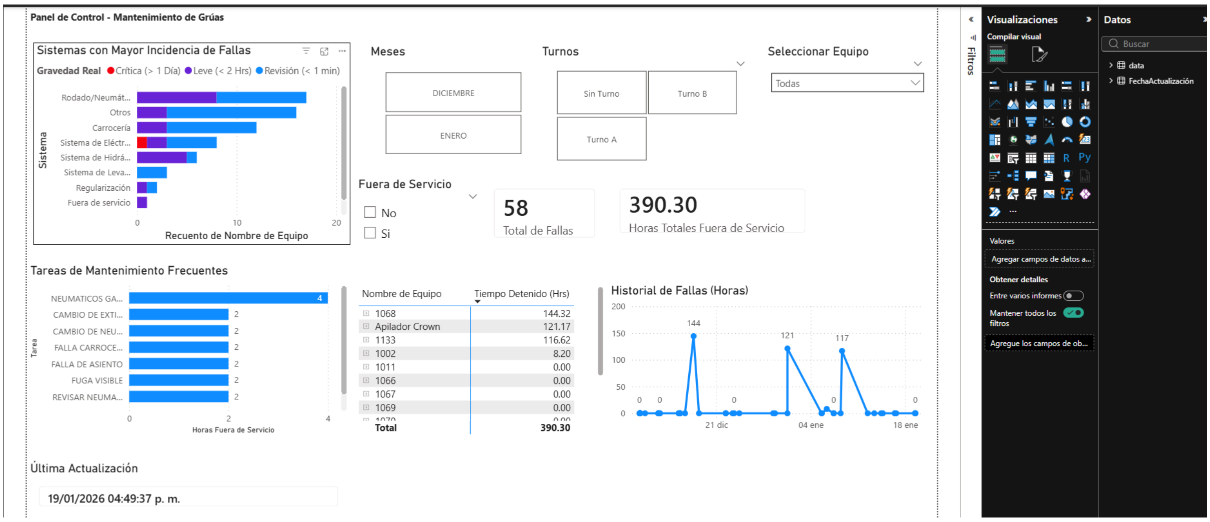

Image 2: Dashboard Panel de Control- Mantenimiento de Grúas.

3. Analysis

3.1 Company Analysis

Jugos Vital SPA exhibits a high production capacity supported by rigorous international quality standards, such as FSSC 22000. The facility functions as a strategic hub for the Coca-Cola System in Chile, maintaining a robust workforce of approximately 251 to 500 employees. This human capital is distributed across administrative, logistical, and operational roles, working in a synchronized 24/7 environment to meet high domestic demand for products like juices, nectars, and sports drinks. The plant's productivity is directly linked to the uptime of its high-tech bottling and flexible lines, which require constant technical supervision to ensure operational continuity.

The analysis conducted during the internship revealed that, despite the high-tech nature of the production equipment, many administrative and maintenance controls still relied on traditional manual processes. A primary example of this was forklift fleet management, where operators were required to fill out a physical form titled Control Operacional de Grúas at the start of every shift. These documents recorded critical safety and mechanical data, such as tire condition, hydraulic leaks, and the status of safety lights. The daily collection and manual review of these physical sheets across the five key areas—production, warehouse, yard, waste management, and formulation—created a significant information gap, as the data was not immediately available for statistical analysis or proactive maintenance planning.

In an environment with high asset utilization, the efficiency of the maintenance of the logistics fleet is vital to prevent bottlenecks in the distribution process. The company is currently in a positive maturation phase toward Industry 4.0, where data integration across departments is the key to maintaining its competitive position. By transitioning from these physical paper records to digital platforms like Fracttal and analytical ecosystems like Power BI, the department can eliminate human error in data entry and gain real-time visibility. This digital transformation underscores the need for continued investment in management tools to maximize operational uptime and resource allocation. The integration of field data into digital technical files allows for a more accurate calculation of indicators such as mean time to resolution and equipment availability, ensuring that Jugos Vital SPA can maintain its high production standards while optimizing the long-term reliability of its industrial infrastructure.

3.2 Work Analysis

The work performed contributed a substantial improvement to the operational visibility and administrative efficiency of the maintenance area. A critical finding during the evaluation was that while digital tools like Power BI provide high strategic value, their effectiveness depends entirely on the quality of the source data. A common error observed during the data collection phase across the five plant areas was the inconsistency in manual checklist entries, which initially complicated the statistical analysis. To address this, it is proposed as a permanent improvement to implement digital tablets in the field to eliminate paper records entirely. This measure would reduce human error during data capture and automate the update of key performance indicators, such as the mean time to repair (MTTR) and overall equipment availability, ensuring that management has access to accurate and immediate operational data at all times.

Beyond data analysis, the logistical management of the technical workforce across three shifts (A, B, and C) highlighted the complexity of maintaining 24/7 operational continuity. Coordinating the transportation for Shift C demonstrated that maintenance support is as much a logistical challenge as a technical one. Furthermore, the integration of SAP for monitoring spare parts provided a vital link between the warehouse and the workshop. By managing procurement orders—including technical specifications, prices, and vendor data—it was possible to reduce delays caused by incorrect components, proving that a well-managed supply chain is essential for minimizing equipment downtime.

Finally, the organizational restructuring through the 5S methodology and the development of Standard Operating Procedures (SOPs) addressed the need for knowledge standardization. The optimization of Excel templates using advanced functions further reduced administrative lead times, allowing for a more agile flow of information. The creation of a photographic inventory complemented these efforts by providing a visual bridge between the digital platform (Fracttal) and the physical assets. Collectively, these interventions moved the department toward a more proactive maintenance culture, where the combination of standardized procedures, logistical precision, and data-driven insights ensures the long-term reliability of the plant's industrial infrastructure.

4. Personal Reflection

The professional internship at Jugos Vital SPA has been a transformative experience that allowed me to apply the analytical rigor of Civil Engineering within a complex industrial environment. Working under the guidance of Matías Valenzuela taught me the critical importance of meticulous planning and the effective management of human resources. Beyond the technical challenges, the human quality of the maintenance team was a fundamental pillar of my development. From the very beginning, I was welcomed into an inclusive environment where I never felt excluded. The openness and willingness of my colleagues to share their knowledge created a safe space where I could express my doubts and ask questions without hesitation, which significantly accelerated my learning curve.

The development of the Power BI dashboard was a significant self-learning challenge that strengthened my ability to solve complex technical problems autonomously. This project demonstrated that engineering principles are universal and can be applied to optimize any system, whether it is a physical structure or a logistical process. From a human perspective, managing transportation logistics and personnel shifts provided a real-world understanding of an engineer's responsibility toward their team. I learned that technical efficiency must be balanced with empathy and clear communication to maintain a productive work environment.

I am leaving Jugos Vital SPA with the satisfaction of having implemented a functional tool that adds genuine value to the company and having been part of a team that values mentorship and professional respect. This experience has significantly broadened my vision of asset management and logistics, providing a solid foundation for my future career as a civil engineer capable of integrating technical expertise with strategic leadership and a deep appreciation for collaborative work environments.